

VLSI Physical Design Project

Implementation Documentation

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Tools used:

Synthesis: Synopsys Design Compiler – V.2023.12-SP4

Place & Route: Synopsys IC Compiler (ICC2) – X-2025.06-SP2

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1. Block Details

Block Name: Tile

No. of ports: 1637

- a. Input ports - 838
- b. Output ports - 797
- c. Inout ports – 0

No. of macros: 18

No. of clocks: 1

Die size: 1163.82 um x 1163.70um

Die Area: 1,354,337.334 um²

Initial core utilization: 70%

2. Design Specifications

2.1 Technology info:

Technology node: 28nm

Standard cell height: 0.9 um

Standard cell VT types: HVT, SVT, LVT

2.2 Clock Specifications

Clock name	Clock period	Master/Generated clock	Sinks
clk	1.70 (588 Mhz)	Master (primary port)	101158

3. Implementation Metrics & Checklist

3.1 Synthesis Stage Metrics

Metric	Value	Unit
RTL files loaded successfully without errors	Pass	Pass/ Fail
Elaborate finished without any errors	Pass	Pass/ Fail
SDC file applied without any errors	Pass	Pass/ Fail
No errors in log file	Pass	Pass/Fail
Applied don't use for LVT, CK*, High drive strength cells	Pass	Pass/Fail
Gate-level netlist generated	Pass	Pass/Fail
Synthesis reports generated	Pass	Pass/Fail
Total Cell Area	945493.838409	μm ²
Combinational Area	197433.682714	μm ²
Sequential Area	208876.999445	μm ²
Macro area	539183.15650	μm ²
Total Cells	324021	count

Sequential Cells	103304	count
Combinational Cells	218119	count
Buf/Inv Cells	26463	count
Worst Setup Slack (WNS)	-0.21	ns
Total Negative Slack (TNS)	240.41	ns
No. of Violating paths	2061	Count
Total Power	286.46	mW
Check timing issues	Clean (3 waivers documented)	Clean/Not clean
Check design issues	clean	Clean/Not clean

Overall Synthesis Stage: Complete

3.2 Floorplan Stage Metrics

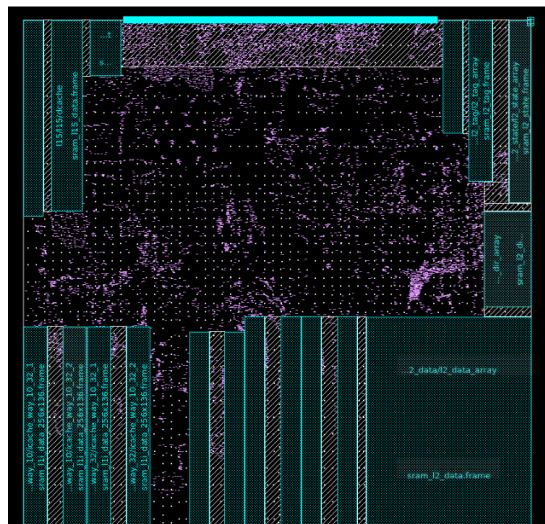
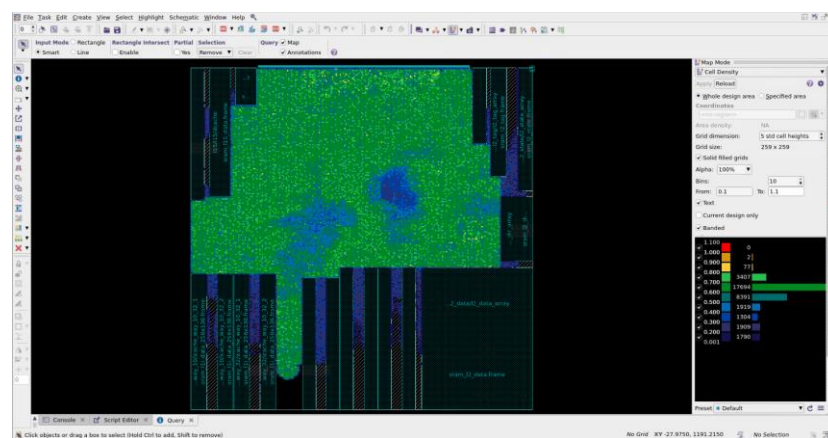
Metric	Value	Unit
Design imported successfully	Pass	Pass/Fail
Core area defined with proper utilization (70%)	Pass	Pass/Fail
No linking issues	Pass	Pass/Fail
Core Width	1162.14	μm
Core Height	1161.9	μm
Core Area	1350290.466	μm ²
Die Width	1163.82	μm
Die Height	1163.7	μm
Die Area	1354337.334	μm ²
Core Utilization	72.63	%
Core offsets	0.84 x 0.9	μm
Aspect Ratio	1.0 (Square)	
Number of Macros	18	count
Macro Area	539183.16	μm ²
Ports placed correctly	Pass	Pass/Fail

Endcap cell issues	Clean	Clean/Not clean
Tap cell issues	Clean	Clean/Not clean
PG opens	0	Count
PG shorts	0	Count
PG connectivity	Clean	Clean/Not clean
PG missing vias	0	Count
Zero interconnect timing (pre place timing)	-0.21/-243.53/2132	WNS/TNS/FEP
Check timing	Clean	Clean/Not clean
No errors in logfiles	Pass	Pass/Fail

Overall Floorplan Stage: Complete/Incomplete

3.3 Placement Stage Metrics

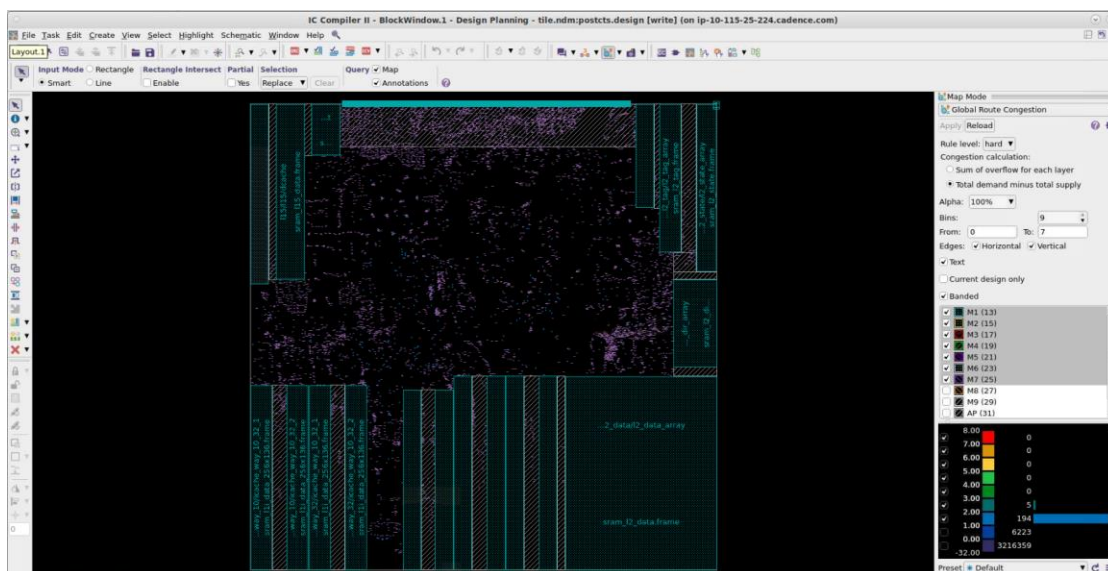
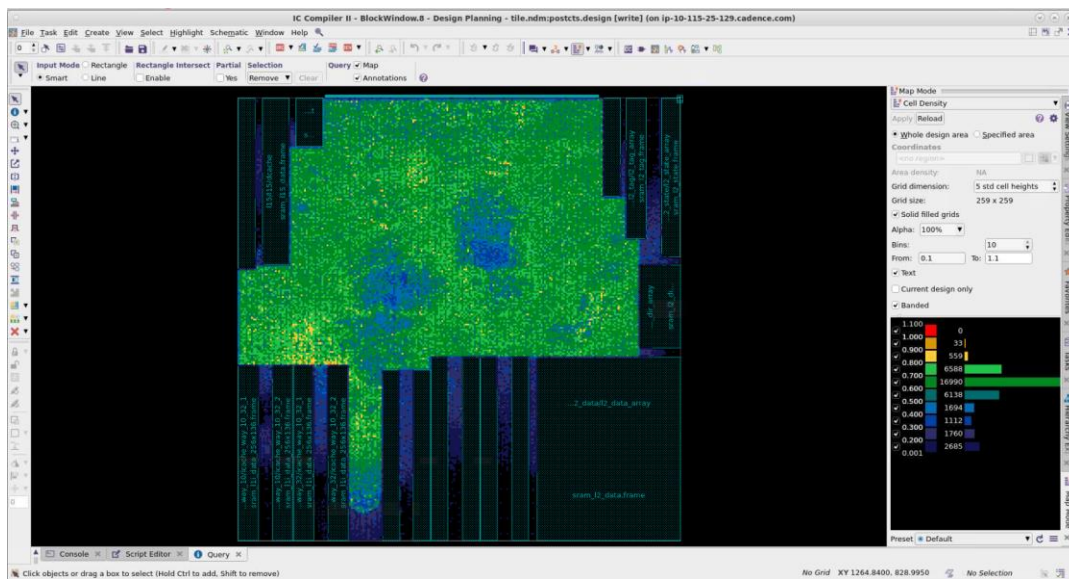
Metric	Value	Unit
core utilization	75.44	%
check_place or check_legality	Clean	Clean/ Not clean
Congestion	0.11% H, 1.18% V	H% & V% Innovus: Max hotspot score
Congestion acceptable	Pass	Pass/Fail
Cell density acceptable	Pass	Pass/Fail
Setup timing (reg2reg)	-0.10/-5.21/152	WNS/TNS/FEP
Setup timing (in2reg)	-0.19/-9.93/192	WNS/TNS/FEP
Setup timing (reg2out)	-0.20/-53.12/784	WNS/TNS/FEP
Setup timing (in2out)	0.00/0.00/0	WNS/TNS/FEP
Max transition violations	0/0	WNS/Count
Max Capacitance violations	0/0	WNS/Count
Spare cells inserted	Yes	Yes/No
No. of spare cells	1360	Count



3.4 CTS Stage Metrics (PostCTS)

Metric	Value	Unit
Clock Buffers Inserted	402	count
Clock Inverters Inserted	212	count
Total Clock Tree Cells	1247	count
Only CK*LVT cells used in clock tree	Fail	Pass/Fail
Clock tree reports generated	Yes	Yes/No
Clock Tree Area	7086.37	μm ²
Global Skew	360 – 650 (worst: ssg0p81vm40c_cw)	ps
Max Clock Latency	480 – 850 (worst: ssg0p81vm40c_cw)	ps
Clock Transition Limit	0.2	ns
Clock fanout limit	1000000 (default)	count
Setup timing (reg2reg)	-0.20/-25.47/511	WNS/TNS/FEP
Setup timing (in2reg)	-0.20/-24.77/626	WNS/TNS/FEP
Setup timing (reg2out)	-0.25/-87.79/785	WNS/TNS/FEP
Setup timing (in2out)	0.00/0.00/0	WNS/TNS/FEP
Hold timing (reg2reg)	-0.11/-7.73/278	WNS/TNS/FEP
Max transition violations	0/0	WNS/Count
Max Cap violations	0/0	WNS/Count
Setup uncertainty	0.255	ns
Hold uncertainty	0.025	ns
Clock net DRCs	21256	Count
core utilization	75.22	%
check_place or check_legality	Clean	Clean/ Not clean
Congestion	0.10% H, 1.00% V	H% & V% Innovus: Max hotspot score

Snapshot of cell density, congestion map.



Overall CTS & PostCTS Stage: Complete

3.5 Routing Stage Metrics

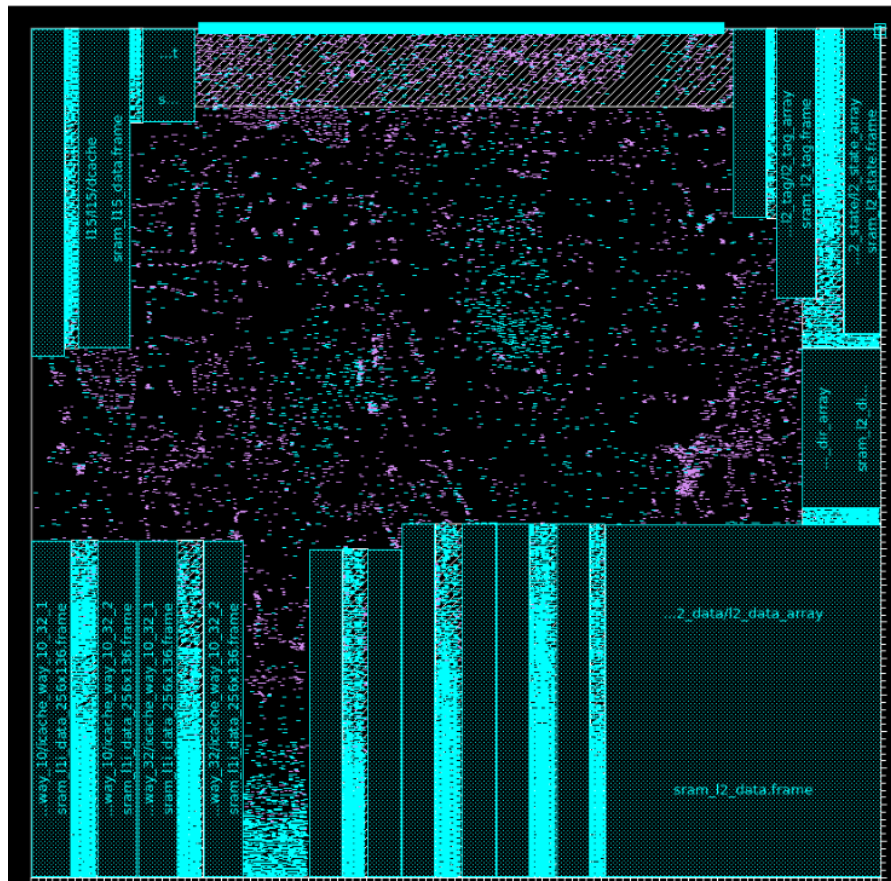
Metric	Value	Unit
Core utilization	75.22	%
DRC violations	38	count
Shorts	6	count
Opens	1	count
Antenna violations	0	count

Overall Route Stage: Complete

3.6 Post-Route Metrics (Final)

Metric	Value	Unit
core utilization	99.97	%
check_place or check_legality	Clean	Clean/ Not clean
DRC violations	0	count
Shorts	0	count
Opens	0	count
Setup timing (reg2reg)	0.0000/0.0000/0	WNS/TNS/FEP
Setup timing (in2reg)	0.0000/0.0000/0	WNS/TNS/FEP
Setup timing (reg2out)	0.0000/0.0000/0	WNS/TNS/FEP
Setup timing (in2out)	0.0000/0.0000/0	WNS/TNS/FEP
Hold timing (reg2reg)	0.0000/0.0000/0	WNS/TNS/FEP
Max transition violations	0.00/0	WNS/Count
Max Cap violations	0.000/1	WNS/Count
Final GDSII generated	Yes	Yes/No
Final DEF generated	Yes	Yes/No
Final netlist generated	Yes	Yes/No
All signoff reports generated	Yes	Yes/No

Snapshot of design with fillers. (Disable all metal layers)



4. Issues and Fixes

Issue Template

Issue #1

Issue Title	Routing Blockage Caused M1 Power Rail Disconnection and M4 Fat-Metal DRC Violations
Design Stage	Floorplan
Issue Category	PG / DRC
Tool	ICC2 X-2025.06-SP2
Severity	High
Issue Description	While running check_pg_drc, the M1 power rail was found to be disconnected due to a routing blockage, preventing VDD/VSS connectivity to the affected tap cells and endcap cells. The same check also reported a large number of insufficient spacing violations on the M4 layer.
Error Message / Log	icc2_shell> check_pg_drc Total number of errors found: 21817 11 illegal overlaps on M1 21806 insufficient spacings on M4
Root Cause Analysis	The routing blockage overlapped the M1 power rail, interrupting power continuity to the affected rows. In addition, the macro OBS regions conflicted with the M4 power stripes, causing the insufficient spacing violations.
Debug Steps Followed	The affected M1 rail was traced to the routing blockage and verified to be disconnected. The M4 DRC report was also reviewed, confirming that all reported violations were insufficient spacing violations.
Solution Applied	The routing blockage was reduced to restore the M1 rail, and the disconnected rail was manually stretched using the Stretch (S) tool. After running connect_pg_net -automatic, the application option plan.pgroute.treat_fat_blockage_as_fat_metal was enabled to resolve the M4 spacing violations. After re-running check_pg_drc, all 11 illegal overlaps were cleared, the M1 power rail connectivity was restored, and the M4 insufficient spacing violations reduced from 21,806 to 0.
Result After Fix	After re-running check_pg_drc, all 11 illegal overlaps were cleared, the M1 power rail connectivity was restored, and the M4 insufficient spacing violations reduced from 21,806 to 0.

Issue #2

Issue Title	Spare Cells Removed During place_opt Due to Optimization
Design Stage	Placement
Issue Category	Flow / Tool behaviour
Tool	ICC2 X-2025.06-SP2
Severity	Medium
Issue Description	After inserting spare cells during placement, all spare cell instances were removed during the place_opt stage and were no longer present in the design.

Error Message / Log	Information: The register PNR_Spare_8_0 is removed because it is unloaded. (SQM-4101) ... Information: 129 registers are removed as unloaded. Use report_transformed_registers for a list. (SQM-3101)
Root Cause Analysis	The spare cells were connected only to tie-high and tie-low cells and had no functional fanout. As a result, place_opt identified them as unused logic and removed them during optimization.
Debug Steps Followed	The place_opt log was reviewed and confirmed that the spare cells were removed during optimization. Their connectivity was also verified, showing that they were connected only to tie cells.
Solution Applied	The placement stage was re-run after reinserting the spare cells and applying the dont_touch attribute before executing place_opt.
Result After Fix	The spare cells were preserved throughout place_opt, and placement completed with clean congestion and legality. The spare cells remained in the design for the rest of the implementation flow.

Issue #3

Issue Title	Tighter CTS Skew Target Degraded Setup Timing
Design Stage	CTS
Issue Category	Timing
Tool	ICC2 X-2025.06-SP2
Severity	Medium
Issue Description	CTS was executed with skew targets of 0.1 ns and 0.05 ns to compare clock tree quality. Although the 0.05 ns run achieved better skew and latency, it resulted in poorer setup timing.
Error Message / Log	No tool error. QoR comparison performed using: • report_qor • skew.rpt
Root Cause Analysis	The tighter skew target required additional clock tree balancing, increasing the number of clock cells and clock latency, which negatively impacted setup timing.
Debug Steps Followed	Compared report_qor and skew.rpt for both CTS runs and verified that the 0.05 ns run inserted more clock tree cells than the 0.1 ns run.
Solution Applied	The 0.05 ns CTS run was discarded, and the 0.1 ns skew target was selected for the remaining implementation stages.
Result After Fix	The 0.1 ns CTS run provided the best overall balance between clock skew and setup timing and was used for the rest of the implementation flow.

Issue #4

Issue Title	Regular AND/OR Gates Found in the Clock Network
Design Stage	Post-Route
Issue Category	Clock Network / Cell Merging
Tool	ICC2 X-2025.06-SP2
Severity	Medium
Issue Description	During post-route clock network review, regular AND and OR gates were found in the clock path instead of dedicated clock network cells.
Error Message / Log	icc2_shell> get_cells -filter "is_clock_network_cell && ref_name =~ AN*" 540 cells returned icc2_shell> get_cells -filter "is_clock_network_cell && ref_name =~ OR*" 93 cells returned
Root Cause Analysis	Fanin/fanout analysis confirmed that these gates were part of the clock-gating logic. During synthesis, a size_cell restriction prevented them from merging into dedicated clock-gating cells.
Debug Steps Followed	Performed fanin/fanout analysis on the identified cells and reviewed the placement and synthesis logs. The synthesis log confirmed that the size_cell restriction prevented cell merging.
Solution Applied	Filtered only the AND/OR gates belonging to the clock network and replaced them using a size_cell loop. The AND gates were swapped to CKAN2D2BWP40P140LVT, while the OR gates were replaced with OR2D2BWP40P140LVT.
Result After Fix	The regular AND/OR gates in the clock network were replaced with the appropriate CK/LVT variants, while the remaining AND/OR gates outside the clock network were left unchanged.

Issue #5

Issue Title	Over-Constrained I/O Delay Caused Setup Timing Violations
Design Stage	Post-Route ECO
Issue Category	Timing / Constraints
Tool	ICC2 X-2025.06-SP2
Severity	High
Issue Description	During setup timing closure, a significant number of interface paths continued to violate setup timing. Running analyze_design_violations -type setup identified external delay as the primary cause.
Error Message / Log	icc2_shell> analyze_design_violations -type setup Root Cause : External Delay
Root Cause Analysis	The input and output delay constraints were set to 70% of the clock period, leaving insufficient timing budget for the internal logic on interface paths.
Debug Steps Followed	Reviewed the output of analyze_design_violations -type setup and confirmed that the setup violations were caused by external delay rather than internal logic.
Solution Applied	Reduced the input and output delay constraints from 70% to 60% of the clock period and re-ran timing analysis.

Result After Fix	The setup violations on the interface paths reduced significantly, with reg-to-out and in-to-reg paths coming close to timing closure. The remaining setup violations were primarily on reg-to-reg paths.
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Issue #6

Issue Title	Inefficient VT Swapping During Setup ECO
Design Stage	Post-Route ECO
Issue Category	Scripting / Timing
Tool	ICC2 X-2025.06-SP2 (TCL)
Severity	High
Issue Description	A TCL script was developed to fix setup violations by swapping cells from HVT to SVT and SVT to LVT. The initial version continued swapping cells even after timing was met, resulting in unnecessary VT swaps.
Error Message / Log	No tool error. The script continued performing VT swaps on paths that had already achieved positive slack.
Root Cause Analysis	The script evaluated path slack only once before performing the VT swaps. Since each swap changes the path timing, the remaining swaps were performed using outdated slack information.
Debug Steps Followed	Compared the setup violations and LVT cell count after each script run, then reviewed the script logic and confirmed that slack was not being updated after every cell swap.
Solution Applied	Modified the script to update the path slack after every VT swap and terminate processing for that path as soon as positive slack was achieved. The script was executed in multiple passes until timing closure.
Result After Fix	Setup violations were reduced to 0 while requiring significantly fewer VT swaps than the initial implementation.

Issue #7

Issue Title	CPPR Disabled by Default Increased Reported timing Violations
Design Stage	Post-Route ECO
Issue Category	Timing
Tool	ICC2 X-2025.06-SP2
Severity	Critical
Issue Description	After completing setup timing closure, hold violations remained difficult to resolve using conventional ECO techniques. While reviewing the timing analysis settings, it was observed that Clock Reconvergence Pessimism Removal (CPPR) was disabled throughout the implementation flow.
Error Message / Log	Before enabling CPPR Clock Reconvergence Pessimism Removal : disabled Hold Violations NUM : 104 WNS : -0.02 TNS : -0.66

	After enabling CPPR icc2_shell> set_app_options \ -name time.remove_clock_reconvergence_pessimism \ -value true Information: Timer using 'SI, Timing Window Analysis, CPPR' Hold Violations NUM : 9 WNS : -0.01 TNS : -0.03
Root Cause Analysis	Although timing derates were already applied, the application option time.remove_clock_reconvergence_pessimism remained disabled. This introduced unnecessary pessimism into the timing analysis, causing a large number of non-genuine hold violations to be reported.
Debug Steps Followed	Reviewed the timing report configuration after setup closure and observed that Clock Reconvergence Pessimism Removal was disabled. Since timing derates were already enabled, this configuration was identified as inconsistent and confirmed before updating the timing analysis settings.
Solution Applied	Enabled time.remove_clock_reconvergence_pessimism and updated timing. The remaining genuine hold violations were then resolved using targeted VT swapping on the launch flip-flops.
Result After Fix	Enabling CPPR reduced the reported hold violations from 104 to 9 immediately. The remaining violations were resolved through targeted VT swaps, achieving complete hold timing closure. The updated timing analysis also improved setup slack, providing additional timing margin during the hold ECO process.

Issue #8

Issue Title	Shorts and DRCs after route_opt — cleared manually and via re-route
Design Stage	Post-Route
Issue Category	DRC
Tool	ICC2 X-2025.06-SP2
Severity	Critical
Issue Description	After route_opt, there were 16 shorts and 13 DRCs in the design.
Error Message / Log	check_routes reported 16 shorts and 13 DRC violations.
Root Cause Analysis	A mix of causes — some were leftover from cell swaps during ECO changing the footprint of cells slightly, and some were general routing congestion in tighter areas of the design.
Debug Steps Followed	Ran check_routes to get the full list of shorts and DRCs. Went through them area by area to separate the ones that could be cleared by re-routing from the ones that needed a manual fix.
Solution Applied	Some of the shorts and DRCs were cleared automatically just by running route_opt again. The remaining ones were fixed manually, one at a time.
Result After Fix	All 16 shorts and all 13 DRCs were cleared to 0.

5. Results and Analysis

5.1 Stage-wise Metric Progression

Stage	Area (μm^2)	Setup WNS (ns)	Setup TNS (ns)	Setup NVP	Power (mW)
Synthesis	945493.838409	-0.21	-240.41	2061	286.43
Placement	958567.23	-0.20	-68.28	1128	279.09
CTS	979860.48	-0.24	-185.45	4712	138.46
Routing	979860.48	-0.25	-332.11	14177	138.86
Post-Route	977693.40	0.00	0.00	0	90.09

5.2 Final Power Breakdown:

Component	Power (mW)	Percentage
Sequential	0.813	0.9%
Combinational	3.37	3.7%
Clock Network	41.3	45.8%
Leakage	0.0961	0.11%
Total	90.10	100%

6. Conclusion and Learnings

6.1 Technical Learnings

Synthesis Learnings:

- I found that some clock-gating logic was not merged into dedicated clock-gating cells during synthesis because of size_cell restrictions. This taught me the importance of verifying the clock network after synthesis rather than assuming all clock-gating logic has been merged correctly.
- I found that some check_timing warnings were automatically resolved during compile_ultra. Instead of trying to fix every initial warning, I learned that it is better to complete the synthesis flow first and then focus only on the warnings and errors that remain.

PnR Learnings:

- I found that reviewing the implementation logs at every stage made debugging much easier. Going back to the synthesis, placement, and timing logs helped me identify the actual source of an issue instead of fixing only the symptoms seen in the current stage.
- I learned that understanding the root cause of a timing violation is more effective than immediately applying ECO fixes. Using timing analysis reports helped me determine whether the issue was caused by constraints, timing settings, or the design itself before making any changes.
- I learned that defining the clock NDR rules before placement produced better results than introducing them later in the flow. This allowed the placer to account for wider clock routes during congestion estimation, leading to improved timing and routing quality.

Verification Learnings:

- I learned that thoroughly reviewing the logs and reports at every implementation stage saved significant debugging effort later in the flow. Verifying the QoR metrics, checking for errors, warnings, and informational messages before moving to the next stage helped identify configuration issues early and prevented them from propagating into later stages.
- I found that physical verification after filler insertion is equally important as timing verification. During chip finishing, I observed that the cell padding applied to the I/O buffers prevented filler cells from being inserted between them. After removing the padding only for the I/O buffers, the fillers were inserted successfully.

Tool Usage Learnings:

- I developed a TCL script to automate setup timing closure by performing incremental VT swapping on violating paths. Instead of swapping every cell on a path, the script recalculates the slack after each swap and stops once timing is met, ensuring that only the minimum number of cells are converted to lower VT cells and avoiding unnecessary power overhead.
- I found that exploring the tool commands and their options made debugging and optimization much easier. Using the -help option helped me understand the available switches for different commands, while report_app_options allowed me to identify application options related to specific optimizations instead of relying on trial and error.
- I found that the default utilization report did not provide the utilization metric I required. This led me to create a custom utilization configuration, where I learned that any object excluded from the utilized area must also be excluded from the available area to obtain a meaningful utilization value. Using the custom core_util configuration produced the expected post-route utilization after filler insertion.

```
create_utilization_configuration core_util -include {hard_macros macro_keepouts}  
report_utilization -config core_util
```

6.2 Final Remarks (to be filled by reviewer)

Reviewer Details:

Name: _____

Date: _____

Comments:
